



BLACK BOX THINKING

Why Most People Never Learn from Their Mistakes—But Some Do

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The Big “So What”

Everyone makes mistakes. It’s how you respond to those mistakes that’ll determine whether you succeed or fail in the long term. Specifically, your ability to learn from failures decides whether you repeat the same errors or use them as building blocks to move toward success. **This book explains how to harness the power of failure to grow, improve and succeed.**

Introduction

 To learn from failures, you need the right systems and mindsets. This is exemplified by the aviation industry and its progressive approach to failures:

- There’re 2 black boxes on an aircraft: one that records mechanical information (e.g. fuel levels), and another that records sounds in the cockpit (including the pilots’ conversations). During accident investigations, these black boxes provide holistic, unbiased information to uncover what happened, so corrective measures can be taken to prevent future accidents.
- Pilots are typically rewarded (not penalized) for reporting mistakes. Accidents are investigated in depth and insights are shared throughout the industry to facilitate learning and improvement.

 In this book, Syed examines case studies from many domains and industries—from healthcare to aviation, non-profits and sports—to show why some people/organizations fail to learn from mistakes while others use them to improve and succeed. In this summary, we’ve organized the key ideas in 2 parts:

KEY QUOTES

“If we wish to fulfill our potential as individuals and organizations, we must redefine failure.”

“Learning from mistakes relies on two components: first, you need to have the right kind of system—one that harnesses errors as a means of driving progress; and second, you need a mindset that enables such a system to flourish.”

- Why we must redefine failures; and
- How to learn from mistakes.



Part 1: Redefining Failures



Why We Avoid and Deny Failure

In any society, there's a tendency to avoid failures, especially when our self-image and professionalism are at stake.



Cognitive dissonance. We feel conflicted when our beliefs or self-image are threatened by our failures. In such cases, we can either (i) accept our mistakes, re-evaluate and adjust our thinking/actions, or (ii) try to protect our ego by denying the error and twisting the facts. Most people tend to adopt the latter. Research shows that we may even erase such failures from our memory. The greater our personal stakes, the more likely we'll experience failure-denial. That is why professionals (whose self-identity and image are strongly tied to their work) often find it hard to admit to their mistakes.

- In 1987, Jimmy Bromgard was convicted of rape and sentenced to prison for 40 years. 13 years later, new DNA tests proved that the crime scene DNA did *not* match Bromgard's, i.e. he had been wrongly convicted. The prosecutor for the case refused to accept the new evidence, since it'd mean that he had made a huge professional mistake in the past. Instead, he came up with many reasons to refute the DNA results, including absurd alternatives (e.g. Bromgard was a chimera with 2 blood types).
- In 2003, the U.S. invaded Iraq on the basis that the latter possessed weapons of mass destruction (WMD). Even when it became clear that there was no WMD in Iraq, more than half the Republicans denied the evidence and insisted that they believed the WMD had been found.

KEY QUOTES

"We cover up mistakes, not only to protect ourselves from others, but to protect us from ourselves."

"The more we have riding on our judgments, the more we are likely to manipulate any new evidence that calls them into question."

"When we are confronted with evidence that challenges our deeply held beliefs we are more likely to reframe the evidence than we are to alter our beliefs."

- In 1954, a cult claimed that the world would end in a flood on 21 Dec 1954, and an alien spaceship would arrive to save believers. Cult members quit their jobs and devoted themselves to spreading the word. When the doomsday hour passed with no flood or spaceship, the members refused to admit their mistake. Instead, they claimed that they had saved the planet with their faith, and went on to recruit even more members.

Such examples may sound ludicrous, yet we don't realize it when we ourselves are thinking and acting in the same way.



A related psychological bias is the confirmation bias: we tend to look for facts that confirm our views and filter out the rest.

- In an experiment, volunteers shared their views on capital punishment before they were given 2 reports—one with strong evidence *for* capital punishment and the other with strong evidence *against* it. After reading the reports, volunteers stuck to their original views—they cited only evidence from the report that supported their views and dismissed the evidence in the other report as flawed.
- The more experience or expertise we have in an area, the more entrenched we become in our beliefs and perspectives, and the more we resist conflicting evidence.



Anticipating and avoiding blame. We're all prone to the fundamental attribution error, i.e. we tend to assign blame to people instead of situational factors. When things go wrong, we're quick to blame others. This keeps us from considering the complexities of a situation, blocks us from identifying the real causes, leads people to hide their mistakes, and hurts those who're unfairly blamed. Everyone loses as a result.



Culture. When cognitive dissonance and blame-avoidance becomes the norm, people don't even *realize* that they're evading their errors. As a result, honest and well-meaning people may end up perpetuating the very problems they seek to solve. For example, in most hospitals, it's taboo for junior staff and nurses to speak up against senior staff and doctors. Errors are rarely disclosed because of (i) a fear of backlash, and (ii) because the medical staff genuinely don't know that they've erred. Medical errors are often

KEY QUOTES

"Intelligence and seniority when allied to cognitive dissonance and ego is one of the most formidable barriers to progress in the world today."

"The most effective cover-ups are perpetrated not by those who are covering their backs, but by those who don't even realize that they have anything to hide."

reframed as “one-off accidents” or “complications” instead of being properly examined. Thus, the same errors get repeated over and over, leading to millions of preventable deaths and tragedies.



Why We Avoid and Deny Failure

Failures tell us that there's a gap between our expectations and reality. They offer valuable information about what went wrong and how we can adjust our assumptions and approach.



There are 2 types of mistakes:

- Mistakes that occur because we don't know the answer (e.g. a pharmaceutical company trying to find a drug for a disease); and
- Mistakes that we make even though we know the answer (e.g. a surgeon forgetting a standard procedure in a surgery).

In either case, the ability to learn from mistakes is crucial for future successes.



Specifically, we can approach failures in 2 ways:

- In closed-loop systems, people do not learn from failures. Data from errors are ignored, and the same mistakes get repeated over and over, often unknowingly.
- In open-loop systems, data from failures are collected, examined and the resulting patterns/insights are translated into practical learning points to facilitate system-wide improvements.



Your approach to failure determines your outcomes. This can be seen in the contrast between the healthcare and aviation industries.

- As of 2013, there was only 1 aviation accident per 2.4 million flights for Western-built jets. On the other hand, there were so many people dying from preventable medical errors that it was equivalent of two B747 aircraft crashing daily.

KEY QUOTES

“Failure is...a signpost. It reveals a feature of our world we hadn't grasped fully and offers vital clues about how to update our models, strategies, and behaviors.”

“When people don't interrogate errors, they sometimes don't even know they have made one.”

- Elaine Bromiley died from brain damage after what was supposed to be a routine operation. When her husband Martin tried to find out what happened, he was told that such inquiries weren't done in healthcare, and Elaine's death was an unexpected complication that couldn't have been avoided. In reality, Elaine had trouble breathing after the anaesthesia was administered. The doctors couldn't diagnose the cause and got so preoccupied with trying to insert a tube down her airways that they didn't hear a nurse's suggestion for a tracheotomy which could've saved Elaine's life. Martin finally secured an investigation which uncovered many areas for improvement, and led to recommendations that helped save many other patients' lives.
- By contrast, the crash of American Airlines' Flight 173 in 1978 was investigated in depth as a norm in the aviation industry. This led to the development of NASA's Crew Resource Management procedure, which was tested, refined, and shared with all airlines to help improve their crew communication and crisis management skills. The number of aircraft accidents fell significantly since the 1970s, and countless lives were saved due to those new processes.

 In reality, failures are essential building blocks to success.

- In any competitive sport or craft, you can only gain mastery through repeated practice. Football legend David Beckham started practicing kick-me-ups since he was 6 years old, trying to keep the ball in the air as long as possible. After 3 years of daily practice (and lots of trial-and-errors), his kick-me-up record increased from 5 to 2003. He then moved on to practicing free kicks, doing tens of thousands of practice kicks (and refining his techniques with each failure) to master his technique.
- However, practice is only effective with instant feedback, e.g. Beckham could only refine his kicks based on the results of his previous attempts. In situations with delayed feedback—e.g. radiologists diagnosing tumors or psychotherapists treating mental illnesses—it's much harder to learn what truly works.

KEY QUOTES

"Learn from the mistakes of others. You can't live long enough to make them all yourself."
—Eleanor Roosevelt

"If a culture is open and honest about mistakes, the entire system can learn from them. That is the way you gain improvements."

"This is the paradox of success: it is built upon failure."

"Feedback, when delayed, is considerably less effective in improving intuitive judgment."

- Science is a great example of a discipline that systematically uses mistakes to make progress. Scientists are constantly testing their theories to confirm what they know, while seeking contradicting info that can disprove their theories. For example, if you only conduct tests at low altitudes, you may conclude that water boils at 100°C. However, the statement “water always boils at 100°C” will be disproved once you conduct tests at a higher altitude.



The good news is, with the right systems and mindset, any person or organization can start to learn from mistakes.

- To reduce preventable medical errors, Dr. Kaplan introduced a Patient Safety Alerts system at Virginia Mason for the hospital staff to report mistakes. Initially, very few reports were made due to the deeply-entrenched culture.
- The wake-up call came when an old lady died in 2004 after being injected with a wrong substance. As people started to report mistakes (and were praised instead of penalized), the culture began to shift. By the time this book was published, about 1,000 alerts were reported each month, allowing errors to be addressed and prevented. For example, texts were added to color-coded wristbands after the crew realized that they didn't work for color-blind nurses. Liability insurance premiums fell 74% and by 2013, Virginia Mason was rated one of the safest hospitals in the world.



Part 2: Learning from Mistakes

Closed loops are usually sustained by (i) people hiding their mistakes, (ii) reframing/justifying their errors, and (iii) interpreting data in a biased way. In this section, we'll explain how to overcome these challenges. Specifically, **black box thinking is about cultivating 2 crucial components to empower people to learn from failure:**

KEY QUOTES

“By finding the places where a theory fails, we set the stage for the creation of a new, more powerful theory.”

- The right systems where mistakes are used to drive progress; and
- The right mindsets to support these systems.



Developing the Right Systems

In complex environments, it's impossible to know everything, and you'll inevitably encounter failures and mistakes. **Instead of pretending that you're infallible or all-knowing, acknowledge your limits and build failures into your processes. Develop sound theories/plans, then test and evolve them using real-world feedback and control groups.**



Cumulative selection is about using trial-and-error to test many alternatives, weed out failed options that don't work and refine viable solutions that can work. To succeed, the lessons and insights from each round of experimentation must be passed on to the subsequent rounds.

- This mirrors Nature's evolutionary process where only the strongest organisms survive to pass on their genes. Unilever used this approach to fix a malfunctioning nozzle in its detergent manufacturing factory. The team mimicked the process of natural selection to randomly test 10 nozzle design variations. They then replicated the best variation for the next batch of 10 design variations. After 45 rounds of tests (and systematically discarding 449 options), they had a significantly superior nozzle design.
- This approach explains why free market economies are so much more efficient than centrally-planned economies. In free markets, only viable businesses survive and their models/ideas are replicated by other companies. This form of "cumulative adaptation" allows only the survival of the fittest.



To succeed, you still need strong theories and concepts. The key is to test your assumptions by seeking real-world feedback, as early and frequently as possible.

KEY QUOTES

"Evolution as a process is powerful because of its cumulative nature."

"Markets work not in spite of the many business failures that occur, but because of them."

- Technologists often use the Lean Startup approach. Instead of trying to build a perfect product from the onset, they create a “minimum viable product” (or MVP), test it with real customers, then use the feedback to adapt the solution until they have a commercially-viable product. This approach combines top-down theories with bottom-up inputs, using an iterative approach to test, learn and improve.
- When Steve Jobs wanted to launch Apple’s retail stores, he started by testing the retail concept in a warehouse. It took several revamps to arrive at a viable concept. He then opened 2 stores to test and refine the concept further. Finally, after rounds of feedback and changes, Apple rolled out its proven retail concept nation-wide.
- When Dropbox founder Drew Houston was developing his cloud storage business concept, he started by releasing an MVP—a video demonstrating how Dropbox would work. This allowed him to test if people were interested in the concept before he even wrote a single line of code. Thereafter, he secured the funds and built the program, using iterations to incorporate users’ feedback.



Data, when analyzed out of context, can be misleading. To ensure that you’re interpreting data meaningfully and accurately, conduct randomized control trials (RCTs) to test the counterfactual.

- For example, if your sales increased after you changed an offer on your website, you may think that the offer worked. In reality, the spike in sales could’ve come from an influencer recommending your product. If you had tested your website across 2 randomized groups—1 with the old offer and 1 with the new offer—you may have found the old offer to generate even more sales.
- Scared Straight was a TV program from the ‘70s that exposed delinquent teens to the reality of prison life, with the goal of scaring them to stay away from crime. For decades, the program was publicized as a success, based on surveys from parents. However, a scientist did a longitudinal study to compare 2 random groups of

KEY QUOTES

“It is by getting the balance right between top-down strategy and a rigorous adaptation process that you hit the target. It is fusing what we already know, and what we can still learn.”

“To leverage the power of failure, you have to be resilient and open.”

delinquent youths—one that went through the program and one that didn't. The results showed that teens who went through Scared Straight were actually 28% more likely to commit crime than those who didn't.

- In reality, most policies and interventions are done without proper tests or control groups. They're often evaluated based on a single set of observational data, leading to biased and inaccurate conclusions. To get a more valid assessment, use RCTs to compare the effects of a change against a control group that's not subject to that change.



However, RCTs are not viable on a large scale where there're too many variables to create an effective control group. In such cases, dissect a complex problem into smaller parts.

- The marginal gains approach is about combining measurements (to assess your current performance) with creative ideation (to discover new ideas for improvement), then applying them in an ongoing loop. Isolate a problem or variable you can test on a small scale, collect data sets and test/iterate the incremental changes. Once you've completed a trial, move on to test other variables, or to test the same variable on a larger scale. Using this approach, you can implement small changes that add up to create massive results.
- For example, to alleviate poverty in Africa, you can break down the problem into smaller sub-components (education, malaria, infrastructure etc.). Then, run RCTs for 1 program at a time. A group of economists wanted to improve children's grades in impoverished parts of Africa. They did a small test by giving free textbooks to 1 group of students in Kenya but not another. When this didn't produce results, they did another test using flipcharts (again with no results). In the end, they discovered a cheap and effective solution: de-worming medication. The medicine removed parasites to boost the children's growth, energy and school attendance by 25%, thereby improving their grades.
- Sir David Brailsford used the marginal gains approach to transform the mediocre British track cycling team into Olympic gold medalists. He achieved this by breaking

KEY QUOTES

"Science is not just about a method...it is also about a mindset."

"Success is a complex interplay between creativity and measurement, the two operating together, the two sides of the optimization loop."

"Marginal gains, as an approach, is about having the intellectual honesty to see where you are going wrong, and delivering improvements as a result."

—Sir David Brailsford

down his big goal into smaller parts. He tested and made incremental changes to many small variables, from the cyclists' bike designs to the mattresses they slept on. These small gains added up to significantly improve the cyclists' overall performance.

- This approach is also adopted by Formula One pit stop teams. They start by developing high-level strategies and identifying the data points needed to test each assumption. Then, they do pit stop practice-runs, examine the data, identify missing data points to be added, and do the practice-runs again in an ongoing loop. This iterative process allows them to test their strategies while collecting multiple sets of data to deepen their understanding of the problems and possible solutions.



Developing the Right Mindsets



Problems and failures are vital ingredients for innovation. The typical creative process goes like this:

- The innovator encounters a failure or problem, which gives him the motivation or reason to innovate.
- He investigates the problem, studying it from multiple angles, exploring and rejecting many options.
- A brilliant solution or epiphany emerges after a gestation period, and usually involves the synthesis of seemingly-separate ideas or technologies. This typically happens in 2 scenarios:
 - (i) He's in a relaxed or "switched off" state (e.g. taking a stroll or shower) instead of focusing narrowly on the problem. In this zoomed-out state, the brain is more likely to make creative associations to see new solutions.
 - (ii) He's facing disagreement, critique, or contradicting info, which forces him to consider the problem from new perspectives. That's why groups with diverse backgrounds tend to be more creative.

KEY QUOTES

"If you break down a big goal into small parts, and then improve on each of them, you will gain a huge increase when you put them all together."

"Removing failure from innovation is like removing oxygen from a fire."

"Without a problem, without a failure, without a flaw, without a frustration, innovation has nothing to latch on to."

*"Creativity is just connecting things."
—Steve Jobs*

- The innovator must now put the idea to the test, going through rounds of iteration (test → fail → refine → test again) until he arrives at a viable solution.



In short, successful innovations don't happen overnight. They take months or years of research, conceptualization, marginal changes, and failures, all of which cumulate in a breakthrough. This requires discipline and persistence.

- James Dyson first conceived the notion of a bagless vacuum cleaner when he was struggling with a vacuum bag choked with dust. He researched and incubated the idea for 3 years before he had an epiphany—to combine a vacuum cleaner with a sawmill. Thereafter, he relentlessly created and tested >5000 prototypes before he arrived a design suitable for commercialization. Dyson certainly wasn't the first to conceive a cyclone vacuum cleaner, but he was the first to persist through many failures to create a workable model.
- Likewise, Pixar doesn't create blockbuster movies like "Toy Story" or "Finding Nemo" by chance. After a storyline is conceived, every frame and element is discussed, critiqued and tested. A typical film goes through >125,000 iterations before the final storyboard is delivered. There are so many changes that the final movie often ends up being totally different from the original concept.



Cultivate a growth mindset, i.e. the belief that people can change and learn from mistakes (as opposed to a fixed mindset that your skills/abilities can't be changed).

- Psychologist Carol Dweck found that children with a growth mindset saw challenges as learning opportunities and were more perseverant compared to those with fixed mindsets. Companies with growth mindsets were more open to collaboration and sharing mistakes, whereas those with fixed mindsets were more preoccupied with hiding mistakes and avoiding blame.
- Willpower and discipline *can* be nurtured. The more you overcome challenges, the easier it becomes. In an experiment, a group of students was given easy tasks that they excelled in, while another group was given harder tasks that they struggled with. When both groups were

KEY QUOTES

"Insight often requires taking a step back and seeing the big picture....But to make a creative insight work requires disciplined focus."

"Winners require innovation and discipline, the imagination to see the big picture and the focus to perceive the very small."

"We progress fastest when we face up to failure—and learn from it."

"If we interpret difficulties as indictments of who we are, rather than as pathways to progress, we will run a mile from failure."

later presented with a tough problem, the 1st group became demoralized quickly while the 2nd group embraced the challenge and even asked for more. Don't allow your self-esteem or the fear of failure to handicap you—instead, use each challenge and failure as an opportunity to grow.



To embrace mistakes, facilitate learning and growth, we must shift away from a culture of assigning/avoiding blame to a culture that's fair and transparent.

- Employees must be able to trust their leaders and organizational systems to evaluate mistakes fairly and transparently. Differentiate between honest mistakes and justifiable blame (e.g. negligence). Give people the room to fail and learn from mistakes.
- Create supporting systems to identify and report errors, collect and analyze data sets, diagnose mistakes and share corrective actions and insights.



Putting it Together



With the right systems and mindset, we can start to transform *any* area of failure into stepping stones for success.

- In the past, the legal and criminal system was considered to be infallible. The discovery of DNA brought new evidence to suggest that hundreds of men had been wrongfully convicted of rape. The Innocence Project was founded to exonerate wrongly-convicted individuals using DNA tests. Initially, they faced strong resistance from judges, prosecutors and legal professionals who had devoted their lives to solving crime and found it hard to face their own mistakes.
- It took years for The Innocence Project to uncover numerous flaws in the legal system and present concrete improvements, ranging from eye-witnessing processes to interrogation techniques and forensic sciences. People finally began to accept that addressing mistakes didn't weaken the legal system. It actually strengthened

KEY QUOTES

"Failure is simply the opportunity to begin again, this time more intelligently."
—Henry Ford

"Systems that do not engage with failure struggle to learn."

the system by ensuring that innocent people weren't wrongly convicted while guilty people didn't walk free.

KEY QUOTES